

# **Digital Circuits**

## **DPP- 3**

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**LEVEL -1**

- 1 The number system with radix 2 is known as
- (a) Binary      (b) Decimal      (c) Octal      (d) Hexadecimal

- 2 A group of four bits is known as
- (a) bit (b) byte (c) nibble (d) word

- 3 The knowledge of binary number system is required for the designers of computer and other digital systems because
- (a) It is easy to learn binary number system
  - (b) It is easy to learn Boolean algebra
  - (c) It is easy to use binary codes
  - (d) The devices used in these systems operates in binary

- 4 The ones complement of the binary number 10001011 is  
(a) 01110100 (b) 11111111 (c) 01110101 (d) 11111110

5 The two's complement of the number 100101000 is

(a) 011010110

(b) 111111111

(c) 011010111

(d) None



- 6 The ones complement of ones complement of the given binary number is  
(a) same binary number      (b) zero      (c) non-zero      (d) none

- 7 The twos complement of the twos complement of the given binary number 1011001 is  
(a) 1011000 (b) 1010110 (c) 1011001 (d) 0100111

- 8 The base or radix a Hexadecimal number system is
- (a) 2                      (b) 8                      (c) 16                      (d) 15

9 The number system with base or radix 8 is known as

- (a) binary      (b) decimal      (c) octal      (d) hexadecimal

10 The decimal equivalent of the binary number 101101 is

(a) 48

(b) 45

(c) 57

(d) 75

11 The decimal equivalent of the binary number 1001.0101 is

(a) 9.125

(b) 9.6125

(c) 9.3125

(d) 9.6215

12 The binary equivalent of decimal number 255 is

- (a) 11111110      (b) 11111101      (c) 11111111      (d) none

13 Identify the binary number represented by the decimal number 10.625

- (a) 1010.011                      (b) 1010.101                      (c) 1010.110                      (d) none



14 The binary equivalent of the decimal number 0.6875 is

(a) 0.1010

(b) 0.1011

(c) 0.1101

(d) 0.0110

15 The octal equivalent of the decimal number 375 is

(a) 560

(b) 567

(c) 565

(d) none

16 The octal equivalent of the decimal number 27.125 is

(a) 33.23

(b) 33.28

(c) 33.1

(d) 33.01

18 The hexadecimal equivalent of the decimal number 375 is

(a) 177

(b) 17A

(c) 1A0

(d) none

18 The hexadecimal equivalent of the decimal number 375 is

(a) 177

(b) 17A

(c) 1A0

(d) none

19 The decimal equivalent of the hexadecimal number “BEED” is

(a) 47877

(b) 48877

(c) 48777

(d) none

20 Encode the decimal number 327.87 in BCD code

(a) 0011 0010 0111. 1000 1001

(b) 0011 0010 0110. 1001 1000

(c) 0011 0010 0111. 1100 1010

(d) none

21 The decimal equivalent of the hexadecimal number 3A.2F is

(a) 58.1835

(b) 58.1385

(c) 58.23

(d) none



- 22 The decimal number 13 is represented in natural BCD as
- (a) 1101                      (b) 0001 0011                      (c) 0000 1101                      (d) none

23 The binary equivalent of the hexadecimal number A0B5 is

(a) 1010 0001 1011 0101

(b) 1010 0000 1010 0101

(c ) 1010 0000 1011 0101

(d) none

24 The octal equivalent of the binary number 11010111 is

(a) 656

(b) 327

(c) 653

(d) D7

25 The decimal equivalent of the binary number 11010111 is

(a) 215

(b) 225

(c) 250

(d) none

26 The octal equivalent of the decimal number 215 is

(a) 327

(b) 372

(c) 237

(d) none

27 The maximum positive and negative numbers which can be represented in twos complement form using  $n$  bits are respectively

(a)  $+(2^{n-1}-1), -(2^{n-1}-1)$

(b)  $+(2^{n-1}-1), -2^{n-1}$

(c)  $+2^{n-1}, -2^{n-1}$

(d)  $+2^{n-1}, -(2^{n-1}+1)$

28 When two  $n$ -bit binary numbers are added then the sum will contain at the most

(a)  $n$ -bit                      (b)  $(n+1)$ -bits                      (c)  $(n+2)$ -bits                      (d)  $(n+n)$ -bits

29 The largest positive number that can be stored in a computer that has 16-bit word length and uses twos complement arithmetic is

(a) 32

(b) 32767

(c) 32768

(d) 65536



30 The maximum positive and negative numbers that can be represented in ones complement using n-bits are respectively

- (a)  $+(2^{n-1}-1)$  and  $-(2^{n-1}-1)$       (b)  $+(2^{n-1}-1)$  and  $-2^{n-1}$   
(c)  $+2^{n-1}$  and  $-(2^{n-1}-1)$       (d) None of these

## KEY

|       |       |       |       |       |       |       |       |       |       |
|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| 01. a | 02. c | 03. d | 04. a | 05. d | 06. a | 07. c | 08. c | 09. c | 10. b |
| 11. c | 12. c | 13. b | 14. b | 15. b | 16. c | 17. a | 18. a | 19. b | 20. a |
| 21. a | 22. b | 23. c | 24. b | 25. a | 26. a | 27. b | 28. b | 29. b | 30. a |

LEVEL -2

1. Convert binary 10101010 to octal.

(a)  $206_8$

(b)  $522_8$

(c)  $255_8$

(d)  $252_8$

**2. Convert octal 377 to binary.**

(a) 11101101

(b) 01111011

(c) 10110111

(d) 11111111

**3. Convert the binary number 0000.1010 to decimal.**

(a) 0.10

(b) 0.55

(c) 0.50

(d) 0.625

4. The decimal equivalent of hexadecimal C9 is \_\_\_\_\_.

(a)  $215_{10}$

(b)  $220_{10}$

(c)  $201_{10}$

(d)  $210_{10}$

**5. Convert the decimal number 12.125 to binary.**

(a) 1100.0110

(b) 1110.0010

(c) 1100.0010

(d) 1010.1100



**6. Convert decimal 213 to binary.**

(a) 11001101

(b) 01111001

(c) 11010101

(d) 11100011

**7. The hexadecimal equivalent of decimal 91 is \_\_\_\_\_.**

(a)  $5B_{16}$

(b)  $4C_{16}$

(c)  $4B_{16}$

(d)  $5D_{16}$

**8. Convert octal 36 to binary.**

(a) 110110

(b) 100110

(c) 110011

(d) 011110

**9. Convert decimal 31 to octal.**

(a)  $37_8$

(b)  $43_8$

(c)  $33_8$

(d)  $30_8$

**10. The decimal equivalent of binary 1010000 is \_\_\_\_\_**

(a)  $96_{10}$

(b)  $80_{10}$

(c)  $78_{10}$

(d)  $84_{10}$

**11. The BCD equivalent of decimal 88 is \_\_\_\_\_**

(a)  $1000\ 1000_{\text{BCD}}$

(b)  $1010\ 1000_{\text{BCD}}$

(c)  $1100\ 0010_{\text{BCD}}$

(d)  $1001\ 0000_{\text{BCD}}$

**12. The hexadecimal equivalent of decimal 241 is \_\_\_\_\_**

(a)  $F1_{16}$

(b)  $D3_{16}$

(c)  $E7_{16}$

(d)  $10A_{16}$

**13. Convert the decimal number 3.375 to binary**

(a) 0101.1100

(b) 0011.1010

(c) 0011.0011

(d) 0011.0110



**14. The hexadecimal equivalent of a binary 0010111101111110 is \_\_\_\_**

(a)  $2F77_{16}$

(b)  $4EEE_{16}$

(c)  $77F2_{16}$

(d)  $2F7E_{16}$

- 15. A signed integer has been stored in a byte using 2's complement format. We wish to store the same integer in 16-bit word. We should copy the original byte to the less significant byte of the word, and fill the more significant byte with**
- (a) 0
  - (b) 1
  - (c) equal to the MSB of the original byte
  - (d) complement of the MSB of the original byte.

**16. Which of the following is 1's complement representation of -34?**

(a) 11011110

(b) 01011101

(c) 11011101

(d) 10100010

**17. Which of the following is the 2's complement form of - 68 ?**

(a) 00111100

(b) 10111100

(c) 01111100

(d) 11000100

**18. The BCD equivalent of the binary number 10101011.101 is**

- (a) 000101110101.011000100101
- (b) 000101110001.011000100101
- (c) 100101110001.011000100101
- (d) 000101110001.011000110101

**19. The 8-bit 1's complement representation of -67 is \_\_\_\_\_**

**20. The 2's complement of a 2's complement binary number is**

- (a) 1's complement of the binary number
- (b) 2's complement of the binary number
- (c) Original number
- (d) None

**21. The 2's complement representation of  $(-89)_{10}$  is**

(a) 10100110

(b) 01011101

(c) 10100010

(d) 10100111



22. The base of the number system for the addition  $34 + 14 = 103$  is

\_\_\_\_\_

**23. Decimal 65 in hexadecimal and BCD number system is respectively**

(a) 41,01010110

(b) 41,01100101

(c) 65,01100101

(d) 14,1110,0101

**24. If  $(10ab0)_2 \times (F)_{16} = (10c001010)_2$  then the value of  $(a + b + c)$  is**

(a) 0

(b) 2

(c) 3

(d) 1

25. If  $(25.5)_8 + (x)_8 = (123.1)_8$  then  $x$  is \_\_\_\_\_

**26. A number is represented in binary system 101011. It's gray code equivalent is**

(a) 111100

(b) 111011

(c) 1010111

(d) 111110

27. A number system is represent by  $(54)_b = (13)_b \times (4)_b$  the base of the number is \_\_\_\_\_?

**28. The addition of two hexadecimal number  $(DADA)_{16}$  and  $(BEA9)_{16}$  is**

(a)  $(DEDA)_{16}$

(b)  $(11001100110000011)_2$

(c)  $(1011100110010011)_2$

(d)  $(26649)_{16}$

- 29. How many one's are present in the binary representation of**  
 **$(8 \times 4096) + (4 \times 256) + (9 \times 16) + 5$**
- (a) 6                      (b) 5                      (c) 3                      (d) 4



**30. The addition of 4 bit two's complement binary number 1111 and 0010 results**

(a) 0001 and an overflow

(b) 1001 and no overflow

(c) 0001 and no overflow

(d) 1001 and an overflow

**31. Consider the addition of number with different bases**

$$(67)_8 + (98)_{10} + (x)_5 + (36)_7 = (241)_9 \text{ the value of } x \text{ is } \underline{\hspace{2cm}}$$

**32. The given hexadecimal number  $(1E.53)_{16}$  is equivalent to \_\_\_\_\_**

- (a)  $(35.684)_8$     (b)  $(36.246)_8$     (c)  $(34.340)_8$     (d)  $(35.599)_8$**

**33. The octal number  $(651.124)_8$  is equivalent to \_\_\_\_\_**

(a)  $(1A9.2A)_{16}$

(b)  $(1B0.10)_{16}$

(c)  $(1A8.A3)_{16}$

(d)  $(1B0.B0)_{16}$

**34. The octal equivalent of the decimal number  $(417)_{10}$  is**

- (a)  $(641)_8$       (b)  $(619)_8$       (c)  $(640)_8$       (d)  $(598)_8$

**35. Convert the hexadecimal number  $(1E2)_{16}$  to decimal:**

(a) 480

(b) 483

(c) 482

(d) 484

**36.  $(170)_{10}$  is equivalent to**

(a)  $(FD)_{16}$

(b)  $(DF)_{16}$

(c)  $(AA)_{16}$

(d)  $(AF)_{16}$

**37. Convert  $(214)_8$  into decimal**

(a)  $(140)_{10}$

(b)  $(141)_{10}$

(c)  $(142)_{10}$

(d)  $(130)_{10}$



**38. Convert  $(0.345)_{10}$  into an octal number**

(a)  $(0.16050)_8$

(b)  $(0.26050)_8$

(c)  $(0.19450)_8$

(d)  $(0.24040)_8$

**39. Convert the binary number  $(01011.1011)_2$  into decimal**

(a)  $(11.6875)_{10}$

(b)  $(11.5874)_{10}$

(c)  $(10.9876)_{10}$

(d)  $(10.7896)_{10}$

**40. Octal to binary conversion :  $(24)_8 = ?$**

(a)  $(111101)_2$

(b)  $(010100)_2$

(c)  $(111100)_2$

(d)  $(101010)_2$

**41. Convert binary to octal :**

**(110110001010)= ?**

- (a)  $(5512)_8$       (b)  $(6612)_8$       (c)  $(4532)_8$       (d)  $(6745)_8$

**42. 1's complement of 1011101 is**

- (a) 0101110      (b) 1001101      (c) 0100010      (d) 1100101

**43. 2's complement of 11001011 is**

- (a) 01010111    (b) 11010100    (c) 00110101    (d) 11100010

44. On subtracting  $(01010)_2$  from  $(11110)_2$  using 1's complement, we get\_\_\_\_\_

(a) 01001

(b) 11010

(c) 10101

(d) 10100

45. On subtracting  $(010110)_2$  from  $(1011001)_2$  using 2's complement, we get \_\_\_\_\_

- (a) 0111001      (b) 1100101      (c) 0110110      (d) 1000011



46. On subtracting  $(001100)_2$  from  $(101001)_2$  using 2's complement, we get

- (a) 1101100      (b) 011101      (c) 11010101      (d) 11010111

**47. On addition of 28 and 18 using 2's complement, we get \_\_\_\_\_**

- (a) 00101110    (b) 0101110    (c) 00101111    (d) 1001111

48. On addition of +38 and -20 using 2's complement, we get \_\_\_\_\_

(a) 11110001

(b) 100001110

(c) 010010

(d) 110101011

**49. On addition of -46 and +28 using 2's complement, we get**

- (a) -10010      (b) -00101      (c) 01011      (d) 0100101

**50. On addition of -33 and -40 using 2's complement, we get \_\_\_\_\_**

- (a) 1001110      (b) -110101      (c) 0110001      (d) -1001001

**51. On subtracting +28 from +29 using 2's complement, we get \_\_\_\_\_**

(a) 11111010

(b) 111111001

(c) 100001

(d) 1

52. The addition of the no.  $(341)_8 + (125)_8 + (472)_8 + (577)_8$  is (        )<sub>8</sub>

53. The addition of the no.  $[4FB]_H + [A12]_H + (C38)_H + [75D]_H$  is (            )<sub>8</sub>



**54. Perform the following operations  $(737)_8 - (123)_{16} + (100)_{10} = ( \quad ? \quad )_{16}$**

55. The radix point  $x$  is \_\_\_\_\_  $(211)_n = (152)_8$

56. The value of  $(241)_8 - (176)_8$  is (        )<sub>16</sub>

57. Perform  $(BC)_{16} - (CB)_{16} = ( \quad )_{10}$

58. The result of  $(27.5)_8 + (74.4)_9$  is (     )<sub>10</sub>

59. Convert  $(432)_5 = (x)_7$

60. Convert  $(A98B)_{12}$  to  $(x)_3$   
The value of x is \_\_\_\_\_

**61. The Binary Representation of the decimal no. 87.375 is**

(a) 1011011.011

(b) 0110111.011

(c) 1010111.101

(d) 1010111.011



62. The Hexadecimal no. CAFE is equivalent to [30223332] in base X. the value of X is \_\_\_\_\_.

63. The Multiplication of Binary no. [ 1011 & 1101] is [ A000B11C] in binary the value of A + B + C in decimal is \_\_\_\_.

**64. The two number represented in signed 2's complement form are  $A = 11111001$  &  $B = 11111000$  . The addition of A & B in 2's complement form is**

- (a) 11110010 with overflow      (b) 11110001 with overflow  
(c) 11110001 without overflow      (d) 11110100 without overflow

**65. Which of the following number having same representation in signed & 2's complement form is**

(a) 1011

(b) 1001

(c) 1010

(d) 1100

66.  $(\text{FE35})_{16}$  XOR  $(\text{CB15})_{16}$  is equal to

- (a)  $(3320)_{16}$       (b)  $(\text{FF35})_{16}$       (c)  $(\text{FF50})_{16}$       (d)  $(3520)_{16}$

67. The octal equivalent of the HEX number AB.CD is **GATE (EC 2009)**

- (a) 253.314      (b) 253.632      (c) 526.314      (d) 526.632

**68** If  $(2.3)_4 + (1.2)_4 = (y)_4$ , then the value of  $y$  is

(a) 10.1

(b) 10.2

(c) 10.01

(d) 1.02

**69. Given  $(135)_x + (144)_x = (323)_x$ . What is the value of  $x$  ?**

(a) 5

(b) 3

(c) 12

(d) 6



70.  $(34.4)_8 \times (23.4)_8$  evaluates to

**GATE (CS 2005)**

- (a)  $(1053.6)_8$     (b)  $(1053.2)_8$     (c)  $(1024.2)_8$     (d) None of these

71. The result of  $(45)_{10} - (45)_{16}$  expressed in 6-bit 2's complement representation is,

**GATE (EC 2010)**

- (a) 011000      (b) 100111      (c) 101000      (d) 101001

72. The two numbers represented in signed 2's complement form are  $P = 11101101$  and  $Q = 11100110$ . If  $Q$  is subtracted from  $P$ , the value obtained in signed 2's complement form is **GATE (ECE 2008)**

(a) 100000111

(b) 00000111

(c) 11111001

(d) 111111001

**73. F's complement of  $(2BFD)_{16}$  is**

(a) E304

(b) D403

(c) D402

(d) C403

74. The Addition of the two numbers  $(1A8)_{16} + (67B)_{16}$  will be

**(IES 2017)**

- (a)  $(889)_{16}$       (b)  $(832)_{16}$       (c)  $(823)_{16}$       (d)  $(723)_{16}$

75. The subtraction of two hexadecimal numbers  $84_{16} - 2A_{16}$  results in **(IES 2016)**

(a)  $2B_{16}$

(b)  $3A_{16}$

(c)  $4B_{16}$

(d)  $5A_{16}$

**76. The addition of 4-bit, two's complement, binary numbers 1101 and 0100 results in** **GATE (CS 2006)**

(a) 0001 and an overflow

(b) 1001 and no overflow

(c) 0001 and no overflow

(d) 1001 and an overflow

77. Let  $A=11111010$  and  $B=00001010$  be two 8 bit 2's complement number . Their product of 2's complement is

**GATE (CS 2004)**

(a) 1100 0100

(b) 1010 0101

(c) 1001 1100

(d) 1101 0101



78. The no. of bytes required to represent the decimal no. 1856357 in packed BCD (Binary Coded Decimal) form is .....

**GATE (ECE 2014)**

**79. The minimum decimal equivalent of the no. (11C.0) is**

(a) 183

(b) 194

(c) 268

(d) 269

80. The hexadecimal representation of  $657_8$  is :

**GATE (CS 2005)**

(a) 1AF

(b) D78

(c) D71

(d) 32F

**81. A number in 4-bit two's complement representation is  $(1001)_2$ . If this number was stored using 1 Byte, then the number will be**

(a) 1001

(b) 11111001

(c) 00001001

(d) none of these

82. A number  $N$  of base  $r$  is represented as  $(N)_r$  Let  $(10)_{16}^3 = (X)_{10}^2$   
then the value of  $X$  is \_\_\_\_\_

83. Consider the addition of numbers with different bases

$$(X)_7 + (Y)_8 + (W)_{10} + (Z)_5 = (K)_9$$

If  $X = 36$ ,  $Y = 67$ ,  $W = 98$  and  $K = 241$  then  $Z$  is \_\_\_\_\_

84. What is the addition of  $(-64)_{10}$  and  $(60)_{16}$  ?

1.  $(42)_8$

2.  $(0110\ 0101)_{\text{excess-3}}$

3.  $(00110100)_{\text{BCD}}$

4.  $(0011\ 0000)_{\text{Gray}}$

(a) only 2

(b) 2 and 4

(c) 2, 3 and 4

(d) 1, 2, 3 and 4

85.  $(B9F.AE)_{16} = (\text{-----})_8$



86.  $(11111000\ 101)_2 = ( ? )_{12}$

(a) 4433

(b) 5654

(c) 1199

(d) 1221

87.  $\sqrt{41} = 5$  is correct in at least one number system \_\_\_\_ is the base of the number system.

88. The minimum decimal equivalent of the number  $(1AC)_x$  is equal to

\_\_\_\_\_

89. Consider the following arithmetic operation

$$(24)_x + (17)_x = (40)_x$$

The least possible value of 'x' for which the above operation is valid is

\_\_\_\_\_

90. Consider the following arithmetic equation  $\frac{302}{20} = 12.1$

The minimum possible non-zero base for the given system is

**91. The number of bytes required to represent a binary 6 bit number in BCD format is equal to \_\_\_\_\_**

92. If  $(211)_x = (152)_8$ , then the value of  $x$  is \_\_\_\_\_

93. Consider the arithmetic operation performed in a particular number system whose radix is equal to 'r'.

$$(23)_r + (44)_r + (14)_r + (32)_r = (223)_r$$

The value of radix 'r' is equal to \_\_\_\_.



**94. Let  $X = (C4)_{16}$ ,  $Y = (32)_{10}$ ,  $Z = (76)_8$  then value of  $X - (Y + Z)$  is**

(a)  $(102)_8$

(b)  $(66)_{10}$

(c)  $(66)_8$

(d)  $(66)_{16}$

**95. 2's complement representation of the following Decimal number -  
(105.75)<sub>10</sub> is**

(a) 1 0 0 1 0 1 1 0.0 1

(b) 1 0 0 1 1 1 1 0.0 1

(c) 1 0 0 0 1 1 1 0.1 0

(d) 1 1 0 0 1 0 1 1.1 0

96. The number of digit 1 present in the binary representation of  $3 \times 512 + 7 \times 64 + 5 \times 8 + 3$  is \_\_\_\_\_.

97. How many 1's are present in the binary Representation of  $(4 \times 1096) + (9 \times 256) + (7 \times 16) + 5$  ? \_\_\_\_\_.

98. Given  $(135)_{\text{base } x} + (144)_{\text{base } x} = (323)_{\text{base } x}$   
What is the value of base  $x$ ? \_\_\_\_\_.

99. Consider the equation  $(123)_5 = (x8)_y$  with  $x$  and  $y$  as unknown. The number of possible solutions is \_\_\_\_\_ . **GATE (2011)**

100. Consider the equation  $(43)_x = (y3)_8$  where  $x$  and  $y$  are unknown. The number of possible solutions is \_\_\_\_\_. **GATE (2015)**

101. A new Binary Coded Pentary (BCP) number system is proposed in which every digit of a base-5 number is represented by its corresponding 3-bit binary code. For example, the base-5 number 24 will be represented by its BCP code 010100. In this numbering system, the BCP code 100010011001 corresponds to the following number in base-5 system

**GATE (2006)**

(A) 423

(B) 1324

(C) 2201

(D) 4231



102. A number in 4-bit two's complement representation is  $X_3 X_2 X_1 X_0$ . This number when stored using 8 – bits will be **GATE (1999)**

(A)  $0000 X_3 X_2 X_1 X_0$

(B)  $1111 X_3 X_2 X_1 X_0$

(C)  $X_3 X_3 X_3 X_3 X_3 X_2 X_1 X_0$

(D)  $\bar{X}_3 \bar{X}_3 \bar{X}_3 \bar{X}_3 X_3 X_2 X_1 X_0$

103. Let  $A = 1111\ 1010$  and  $B = 0000\ 1010$  be two 8-bit 2's complement numbers. Their product in 2's complement is **GATE (2004)**

(A) 1100 0100

(B) 1001 1100

(C) 1010 0101

(D) 1101 0101

**104. Assuming all numbers are in 2's complement representation, which of the following numbers is divisible by 11111011? GATE (2003)**

(A) 11100111

(B) 11100100

(C) 11010111

(D) 11011011

105. The 16-bit 2's complement representation of an integer is 1111 1111 1111 0101; its decimal representation is **GATE (2016)**

106. Let  $X$  be the number of distinct 16-bit integers in 2's complement representation. Let  $Y$  be the number of distinct 16-bit integers in sign magnitude representation. Then  $X - Y$  is \_\_\_\_

**GATE (2016)**

107. Two 2's complement numbers having sign bits 'x' and 'y' are added and the sign bit of the result is 'z'. Then, the occurrence of overflow is indicated by the Boolean function **GATE (1998)**

- (A)  $x y z$                       (B)  $\bar{x} \bar{y} \bar{z}$                       (C)  $\bar{x} \bar{y} z + x y \bar{z}$                       (D)  $xy + yz + zx$

108. When two 8-bit numbers  $A_7 \dots A_0$  and  $B_7 \dots B_0$  in 2's complement representation (with  $A_0$  and  $B_0$  as the least significant bits) are added using a ripple carry adder, the sum bits obtained are  $S_7 \dots S_0$  and the carry bits are  $C_7 \dots C_0$ . An overflow is said to have occurred if

**GATE (2017)**

- (A) The carry bit  $C_7$  is 1
- (B) All the carry bits  $\{C_7 \dots C_0\}$  are 1
- (C)  $(A_7 \cdot B_7 \cdot \bar{S}_7 + \bar{A}_7 \cdot \bar{B}_7 \cdot S_7)$  is 1
- (D)  $(A_0 \cdot B_0 \cdot \bar{S}_0 + \bar{A}_0 \cdot \bar{B}_0 \cdot S_0)$  is 1

**109. Assuming all numbers in 2's complement form, which of the following additions will result in overflow?**

(A)  $1100 + 1100$

(B)  $0011 + 0111$

(C)  $1111 + 0111$

(D) All of the above



**110. If  $73_x$  (in base - x number system) is equal to  $54_y$  (in base - y number system), the possible values of x and y are**

**(A) 8, 16**

**(B) 10, 12**

**(C) 9, 13**

**(D) 8, 11**

**111. An 8-bit 2's complement representation of an integer is FA (Hex). Its decimal equivalent is**

(A) -10

(B) - 6

(C) + 6

(D) +10

**112. Which of the following is a self-complementing code ?**

(A) 8421 code

(B) Excess 3 code

(C) Pure binary code

(D) Gray code

**113. Consider the following multiplication :**

$$(10w1z)_2 \times (15)_{10} = (y01011001)_2$$

**Which one of the following gives appropriate values of w, y and z?**

(A)  $w = 0, y = 0, z = 1$

(B)  $w = 0, y = 1, z = 1$

(C)  $w = 1, y = 1, z = 1$

(D)  $w = 1, y = 1, z = 0$

114.  $(98)_{10} + (1D)_{16} + (72)_8 + (1011)_2 =$

(A)  $(314)_8$

(B)  $(B4)_{16}$

(C)  $(194)_{10}$

(D) None of these

115. A number system is represent by  $(54)_b = (13)_b \times (4)_b$  the base of the number is \_\_\_\_\_?

**116. The base of the number system for the addition  $34 + 14 = 103$  is \_\_\_\_.**

**117. Let  $x_1$  be the maximum value that can be represented in signed 2's complement form using 6 bits.**

**Let  $x_2$  be the minimum value that can be represented in signed 1's complement form using 5 bits.**

**Let  $x_3$  be the minimum value that can be represented in signed 2's complement form using 7 bits.**

**Then find the value of  $x_1 + \frac{x_2}{2} + 2x_3$ .**

(a) -180.5

(b) -104.5

(c) 108.5

(d) 130.5



**118. The 4-bit binary number 1110 represents**

- (a)  $(-1)_{10}$  in signed magnitude system and  $(-2)_{10}$  in signed 2's complement system
- (b)  $(-1)_{10}$  in signed 1's complement system and  $(-2)_{10}$  in signed 2's complement system
- (c)  $(-2)_{10}$  in signed magnitude system and  $(-1)_{10}$  in signed 2's complement system
- (d)  $(-6)_{10}$  in signed magnitude system and  $(-1)_{10}$  in signed 2's complement system

**119. If  $(7.FD6)_{16} = (7.7726)_x$  and  $(7864)_{10} = (1EB8)_y$ , Then the values of x and y are respectively.**

- (a) 4 and 8      (b) 10 and 16      (c) 8 and 16      (d) 12 and 10

120. Which of the following is an invalid state in an 8-4-2-1 Binary Coded Decimal counter ?

(A) 1 0 0 0

(B) 1 0 0 1

(C) 0 0 1 1

(D) 1 1 0 0

121. The number of bytes required to represent the decimal number 1856357 in packed BCD (Binary Coded Decimal) form is \_\_\_\_\_.

122. A number in 4-bit two's complement representation is  $X_3 X_2 X_1 X_0$ . This number when stored using 8-bits will be

(A)  $0000 X_3 X_2 X_1 X_0$

(B)  $1111 X_3 X_2 X_1 X_0$

(C)  $X_3 X_3 X_3 X_3 X_3 X_2 X_1 X_0$

(D)  $\overline{X_3} \overline{X_3} \overline{X_3} \overline{X_3} \overline{X_3} X_2 X_1 X_0$

123. Consider the number given by the decimal expression:

$$16^3 * 9 + 16^2 * 7 + 16 * 5 + 3$$

The number of 1's in the unsigned binary representation of the number is \_\_\_\_\_.

*[1991, 1 Mark]*

123. Convert the following numbers in the given bases into their equivalents in the desired bases.

$$(a) \ 110.101_2 = x_{10} \quad (b) \ 1118_{10} = y_H$$

*[1993, 2 Marks]*

124. Consider n-bit (including sign bit) 2's complement representation of integer number. The range of integer values, N, that can be represented is \_\_\_\_\_  $\leq N \leq$  \_\_\_\_\_

*[1994, 2 Marks]*



125. The number of 1's in the binary representation of  $(3 * 4096 + 15 * 256 + 5 * 16 + 3)$  are:

- (A) 8      (B) 9      (C) 10      (D) 12

*[1995, 2 Marks]*

126. Zero has two representations in

- (A) Sign magnitude      (B) 1's complement  
(C) 2's complement      (D) None of the above

*[1999, 2 Marks]*

127. The number 43 in 2's complement representation is\_\_\_\_\_.

- (A) 01010101                      (B) 1101010  
(C) 00101011                      (D) 10101011

*[2000, 1 Mark]*

128. The 2's complement representation of  $(-539)_{10}$

in hexadecimal is

(A) ABE (B) DBC (C) DE5 (D) 9E7

*[2001, 2 Marks]*

129. In 2's complement addition, overflow
- (A) is flagged whenever there is carry from sign bit addition
  - (B) cannot occur when a positive value is added to a negative value
  - (C) is flagged when the carries from sign bit and previous bit match
  - (D) None of the above

*[2002, 1 Mark]*

130. The 2's complement representation of the decimal value  $-15$  is

- |            |           |
|------------|-----------|
| (A) 1111   | (B) 11111 |
| (C) 111111 | (D) 10001 |

*[2002, 1 Mark]*

131. The decimal value 0.25
- (A) is equivalent to the binary value 0.1
  - (B) is equivalent to the binary value 0.01
  - (C) is equivalent to the binary value 0.00111...
  - (D) cannot be represented precisely in binary

*[2002, 1 Mark]*

132. Assuming all numbers are in 2's complement representation, which of the following numbers is divisible by 11111011?

- (A) 11100111                      (B) 11100100  
(C) 11010111                      (D) 11011011

*[2003, 1 Mark]*



133. If  $73_x$  (in base-x number system) is equal to  $54_y$  (in base y-number system), the possible values of x and y are

(A) 8, 16

(B) 10, 12

(C) 9, 13

(D) 8, 11

*[2004, 1 Mark]*

134. Let  $A = 1111\ 1010$  and  $B = 0000\ 1010$  be two 8-bit 2's complement numbers. Their product in 2's complement is

- |               |               |
|---------------|---------------|
| (A) 1100 0100 | (B) 1001 1100 |
| (C) 1010 0101 | (D) 1101 0101 |

*[2004, 2 Marks]*

135. The hexadecimal representation of  $657_8$  is:

- (A) 1AF    (B) D78    (C) D71    (D) 32F

*[2005, 1 Mark]*

136. The range of integers that can be represented by an  $n$  bit 2's complement number system is:

(A)  $-2^{n-1}$  to  $(2^{n-1} - 1)$

(B)  $-(2^{n-1} - 1)$  to  $(2^{n-1} - 1)$

(C)  $-2^{n-1}$  to  $2^{n-1}$

(D)  $-(2^{n-1} + 1)$  to  $(2^{n-1} - 1)$

*[2005, 1 Mark]*

137. Consider numbers represented in 4-bit gray code. Let  $h_3h_2h_1h_0$  be the gray code representation of a number  $n$  and let  $g_3g_2g_1g_0$  be the gray code of  $(n+1)$  (modulo 16) value of the number. Which one of the following functions is correct?

(A)  $g_0(h_3h_2h_1h_0) = \sum(1, 2, 3, 6, 10, 13, 14, 15)$

(B)  $g_1(h_3h_2h_1h_0) = \sum(4, 9, 10, 11, 12, 13, 14, 15)$

(C)  $g_2(h_3h_2h_1h_0) = \sum(2, 4, 5, 6, 7, 12, 13, 15)$

(D)  $g_3(h_3h_2h_1h_0) = \sum(0, 1, 6, 7, 10, 11, 12, 13)$

*[2006, 2 Marks]*

138. Let  $r$  denote number system radix. The only value(s) of  $r$  that satisfy the equation

$$\sqrt{121_r} = 11_r \text{ is / are}$$

- (A) decimal 10                      (B) decimal 11  
(C) decimal 10 and 11   (D) any value  $>2$

*[2008, 1 Mark]*

139.  $(1217)_8$  is equivalent to

(A)  $(1217)_{16}$

(B)  $(028F)_{16}$

(C)  $(2297)_{10}$

(D)  $(0B17)_{16}$

*[2009, 1 Mark]*

140. P is a 16-bit signed integer. The 2's complement representation of P is  $(F87B)_{16}$ . The 2's complement representation of  $8 \cdot P$  is

- (A)  $(C3D8)_{16}$                       (B)  $(187B)_{16}$   
(C)  $(F878)_{16}$                       (D)  $(987B)_{16}$

*[2010, 1 Mark]*



141. The smallest integer that can be represented by an 8-bit number in 2's complement form is
- (A)  $-256$  (B)  $-128$  (C)  $-127$  (D)  $0$

*[2013, 1 Mark]*

142. The base (or radix) of the number system such that the following equation holds is\_\_\_\_\_.

$$\frac{312}{20} = 13.1$$

*[2014, 1 Mark]*

143. Consider the equation  $(43)_x = (y3)_8$  where  $x$  and  $y$  are unknown. The number of possible solution is\_\_\_\_\_.

*[2015, 2 Marks]*

144. The 16-bit 2's complement representation of an integer is 1111 1111 1111 0101; its decimal representation is \_\_\_\_\_.

*[2016, 1 Mark]*

145. Let  $X$  be the number of distinct 16-bit integers in 2's complement representation. Let  $Y$  be the number of distinct 16-bit integers in sign magnitude representation.
- Then  $X - Y$  is \_\_\_\_\_.

*[2016, 1 Mark]*

146. The representation of the value of a 16-bit unsigned integer X in hexadecimal number system is BCA9. The representation of the value of X in octal number system is

- (A) 571244                      (B) 736251  
(C) 571247                      (D) 136251

*[2017, SET-II, 1 Mark]*

147. Consider the unsigned 8-bit fixed point binary number representation below.

$$b_7 b_6 b_5 b_4 b_3.b_2 b_1 b_0$$

Where the position of the binary point is between  $b_3$  and  $b_2$ . Assume  $b_7$  is the most significant bit. Some of the decimal numbers listed below **cannot** be represented **exactly** in the above representation:

- i. 31.500                      ii. 0.875
- iii. 12.100                    iv. 3.001

- Which one of the following statements is true?
- (A) None of (i). (ii). (iii). (iv) can be exactly represented
  - (B) Only (ii) cannot be exactly represented
  - (C) Only (iii) and (iv) cannot be exactly represented
  - (D) Only (i) and (ii) cannot be exactly represented

148. Consider  $Z = X - Y$ , where  $X$ ,  $Y$  and  $Z$  are all in sign-magnitude form.  $X$  and  $Y$  are each represented in  $n$  bits. To avoid overflow, the representation of  $Z$  would require a minimum of:

- (A)  $n$  bits                      (B)  $n + 1$  bits  
(C)  $n + 2$  bits                (D)  $n - 1$  bits

*[2019, 1 Marks]*



149. In 16-bit 2's complement representation, the decimal number  $-28$  is:

(A) 1000 0000 1110 0100

(B) 0000 0000 1110 0100

(C) 1111 1111 0001 1100

(D) 1111 1111 1110 0100

*[2019, 1 Mark]*

149. Two numbers are chosen independently and uniformly at random from the set  $\{1, 2, \dots, 13\}$ . The probability (rounded off to 3 decimal places) that their 4-bit (unsigned) binary representations have the same most significant bit is \_\_\_\_\_.

150. Consider the equation  $(123)_5 = (x8)_y$  with  $x$  and  $y$  as unknown. The number of possible solutions is \_\_\_\_\_ .

*[2014, 1 Mark]*